

All My Work

Your saved learning record for Molecular Genetics.

1 completed activity run. Required checks and formative practice are recorded separately.

Unfinished work: 0 required check(s), 1 practice run(s). Resume them from their original sections; they are not counted as completed.

Writing and flash-card self-assessments are not automatically graded. Completion is not a mastery certificate.

Mixed practice

9/17/2026, 8:43:30 PM • Elapsed 0m 0s • Formative practice

Review submitted responses and attempts

First-try result: 6/6. Later corrections are recorded separately.

Put these steps in order: Translation and the genetic code

Options shown: A matching tRNA supplies an amino acid • A stop codon triggers chain release • Ribosome advances along mRNA • Ribosome establishes the reading frame • Peptide bonds extend the chain

Attempt 1: Ribosome establishes the reading frame → A matching tRNA supplies an amino acid → Peptide bonds extend the chain → Ribosome advances along mRNA → A stop codon triggers chain release — Correct.

Reference: Ribosome establishes the reading frame → A matching tRNA supplies an amino acid → Peptide bonds extend the chain → Ribosome advances along mRNA → A stop codon triggers chain release

Explain restriction enzyme.

Self-report: Study again.

Reference: An enzyme that cuts DNA at particular recognition sequences.

Which term matches this description? The DNA strand read by RNA polymerase to construct complementary RNA.

Options shown: coding strand · RNA polymerase · template strand · RNA splicing

Attempt 1: template strand — Correct.

Reference: template strand

Identify the term illustrated: Coding DNA 5'-ATG-3' corresponds to mRNA 5'-AUG-3'.

Attempt 1: coding strand — Correct.

Reference: coding strand

Select the TWO accurate statements about solving dna, rna and protein problems.

Options shown: If a sequence has the right length, its bases and direction must be correct. · Complementing letters without tracking direction can reverse the message. · An amino-acid sequence is a series of RNA bases. · A practice fragment must state whether it is a complete message or an internal segment.

Attempt 1: true-1 → true-2 — Correct.

Reference: Complementing letters without tracking direction can reverse the message. · A practice fragment must state whether it is a complete message or an internal segment.

What does letter A identify?

Options shown: Loading wells · Larger fragments · Smaller fragments · DNA migration toward positive electrode

Attempt 1: Loading wells — Correct.

Reference: Loading wells

Which term fits this definition? A heritable change associated with a cell taking up external genetic material.

Self-report: Study again.

Reference: transformation

A cell increases production of an enzyme without a DNA sequence change. What process most directly explains this?

Self-report: Study again.

Reference: Altered expression of the relevant gene. A change in gene activity can change protein output without a mutation.

Correct this incorrect statement: An amino-acid sequence is a series of RNA bases.

Self-report: Study again.

Reference: The ordered amino acids in a peptide or protein. It uses amino-acid names rather than 5' and 3' nucleotide ends.

A genetic intervention changes a target sequence but the intended biological function remains unmeasured. What is established?

Options shown: A guaranteed complete cure · No genetic change of any kind · The sequence change, not yet the intended functional benefit · That every tissue has the same response

Attempt 1: The sequence change, not yet the intended functional benefit — Correct.

Reference: The sequence change, not yet the intended functional benefit

Frayer work

transformation — Draft

Meaning: An edited explanation.

Key features: A specific learner explanation 2.

Example: A specific learner explanation 3.

Non-example / common confusion: A specific learner explanation 4.

hereditary material — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

radioactive tracer — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

controlled comparison — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

nucleotide — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

complementary base pairing — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

antiparallel — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

Chargaff's relationships — Draft

Meaning: (not yet written)

Key features: (not yet written)

Example: (not yet written)

Non-example / common confusion: (not yet written)

Collected optional responses

Biotechnology and unintended consequences

The comparison alone cannot isolate the proposed chemical effect. Hormone production and receptor response are distinct steps.

Textbook question work

Draft and saved question responses. Optional and ungraded.

No textbook question work yet.

Supported and independent practice